
Retinal Disease Detector

*A Deep Learning System for Diabetic Retinopathy
Severity Classification from Fundus Images*

Independent Project Report

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Abstract

Diabetic retinopathy (*DR*) is a leading cause of preventable blindness. Early detection through regular fundus screening significantly reduces the risk of vision loss, yet manual grading is time-consuming and requires specialised expertise often unavailable in low-resource settings. This report presents the design, implementation, and evaluation of an end-to-end deep learning system for automated *DR* severity classification from retinal fundus photographs.

The system employs a fine-tuned EfficientNet-B3 convolutional neural network to classify images into five severity grades (0–4) according to the International Clinical Diabetic Retinopathy severity scale. A clinically motivated preprocessing pipeline based on Contrast Limited Adaptive Histogram Equalisation (*CLAHE*) is applied to enhance retinal features. The model was trained and validated on the APTOS 2019 Blindness Detection dataset comprising 3,662 fundus images split into 2,929 training and 733 validation samples (80/20 stratified).

The model achieved a Quadratic Weighted Kappa (*QWK*) of **0.9053** and an overall accuracy of **83.6%** on the validation set. Per-class analysis revealed near-perfect performance on Grade 0 ($F1 = 0.98$) and strong performance on Grade 2 ($F1 = 0.80$), while Grade 3 (Severe *NPDR*) remained challenging ($F1 = 0.36$) due to class imbalance. The system is deployed as a containerised Flask web application with a public interface and includes Grad-CAM based explainability. The system is a research prototype and is **not** a medical device; clinical validation and regulatory approval would be required before any real-world diagnostic use.

Keywords: diabetic retinopathy, deep learning, EfficientNet-B3, fundus imaging, transfer learning, Grad-CAM, explainable AI, class imbalance.

Nomenclature and Acronyms

This section defines the abbreviations and symbols used throughout the report. Acronyms are listed alphabetically; symbols are grouped by category.

Acronyms

Acronym	Meaning
AI	Artificial Intelligence — the broad field of computer systems that perform tasks normally requiring human intelligence.
APTOS	Asia Pacific Tele-Ophthalmology Society — organiser of the 2019 Kaggle diabetic retinopathy competition whose dataset was used here.
AUC	Area Under the Curve — a single number summarising the performance of a classifier across all decision thresholds.
BME	Biomedical Engineering — the application of engineering principles to medicine and biology.
CAD	Computer-Aided Diagnosis — software that assists clinicians in interpreting medical data.
CLAHE	Contrast Limited Adaptive Histogram Equalisation — a local image-enhancement technique that boosts the visibility of small features such as microaneurysms without over-amplifying noise.
CNN	Convolutional Neural Network — a class of deep neural network especially suited to image analysis.
CPU	Central Processing Unit — the general-purpose processor of a computer.
DR	Diabetic Retinopathy — a microvascular complication of diabetes that damages the retina and can lead to blindness.
F1-score	Harmonic mean of precision and recall — a single balanced measure of classification quality.
FN	False Negative — a positive case that the model incorrectly classified as negative.
FP	False Positive — a negative case that the model incorrectly classified as positive.

Acronym	Meaning
GPU	Graphics Processing Unit — a processor optimised for the parallel arithmetic used in deep learning.
Grad-CAM	Gradient-weighted Class Activation Mapping — a technique that produces a heatmap showing which regions of an image most influenced a model’s prediction.
IRMA	Intraretinal Microvascular Abnormalities — abnormal vessel formations visible in severe diabetic retinopathy.
JSON	JavaScript Object Notation — a lightweight, human-readable data format used by the prediction API.
LAB	A colour space with three channels: Lightness (L), and two colour-opponent channels (a, b). Useful because contrast can be adjusted on the L channel alone.
LFS	Large File Storage — a Git extension for tracking large binary files such as model weights.
LR	Learning Rate — the step size used by the optimiser when updating network weights during training.
NPDR	Non-Proliferative Diabetic Retinopathy — the earlier stages of DR (grades 1–3) where no new vessels have formed.
OCT	Optical Coherence Tomography — a non-invasive imaging method that produces cross-sectional views of the retina.
PDR	Proliferative Diabetic Retinopathy — the most severe grade (4), characterised by abnormal new vessel growth.
PNG	Portable Network Graphics — a lossless image file format.
QWK	Quadratic Weighted Kappa — a statistical measure of agreement between predicted and true ordinal labels, penalising large errors more heavily than small ones.
ReLU	Rectified Linear Unit — a simple non-linear activation function: $\max(0, x)$.
RGB	Red, Green, Blue — the standard additive colour model for digital images.
ROC	Receiver Operating Characteristic — a curve plotting true-positive rate against false-positive rate across thresholds.
SDK	Software Development Kit — a collection of tools and libraries for building applications on a platform.
TP	True Positive — a positive case that the model correctly classified as positive.

Acronym	Meaning
VEGF	Vascular Endothelial Growth Factor — a signalling protein that promotes the growth of new blood vessels and drives proliferative DR.
WHO	World Health Organization.

Symbols

Symbol	Meaning
κ	Kappa — the agreement coefficient (see QWK).
α_k^c	Grad-CAM weight for feature map k and class c .
A^k	The k -th feature map of the last convolutional layer.
$E_{i,j}$	Expected agreement in cell (i, j) of the confusion matrix under independence.
F_1	F1-score.
$L_{\text{Grad-CAM}}^c$	Grad-CAM heatmap for class c .
N	Number of severity grades (5 in this project).
$O_{i,j}$	Observed count in cell (i, j) of the confusion matrix.
P	Precision.
R	Recall.
$w_{i,j}$	Quadratic penalty applied to cell (i, j) of the confusion matrix.
$\mathbf{W}_1, \mathbf{W}_2$	Weight matrices of the classification head.
$\mathbf{x}$	Input image tensor.
$\hat{\mathbf{y}}$	Predicted class-score vector.
$\mathbf{z}$	Feature vector extracted by the EfficientNet-B3 backbone.
μ, σ	Mean and standard deviation used for image normalisation.

Contents

Abstract	1
Nomenclature and Acronyms	2
1 Introduction	10
1.1 Clinical Motivation	10
1.2 The Screening Challenge	10
1.3 Problem Statement	10
1.4 Objectives	10
1.5 Contributions	11
1.6 Report Structure	11
2 Clinical and Technical Background	12
2.1 Diabetic Retinopathy: Pathophysiology	12
2.2 International Clinical Severity Scale	12
2.3 Fundus Imaging	12
2.4 Machine Learning for Medical Imaging	13
3 Literature Review	14
3.1 Early CAD Systems	14
3.2 Deep Learning Era	14
3.3 APTOS 2019 Competition	14
3.4 Explainability in Medical AI	14
4 Dataset	15
4.1 APTOS 2019 Blindness Detection	15
4.2 Class Distribution	15
4.3 Train/Validation Split	15
5 Methodology	17
5.1 System Overview	17
5.2 Preprocessing Pipeline	18
5.2.1 Rationale for CLAHE	18
5.2.2 Data Augmentation	18
5.3 Model Architecture	18
5.4 Training Procedure	19

5.5	Evaluation Metrics	19
5.5.1	Quadratic Weighted Kappa	19
5.5.2	Accuracy, Precision, Recall, F1	19
5.6	Explainability: Grad-CAM	20
6	Implementation and Deployment	21
6.1	Software Stack	21
6.2	Web Application	21
6.3	Docker Container	22
6.4	Deployment	22
7	Results	23
7.1	Overall Performance	23
7.2	Training History	23
7.3	Per-Class Performance	24
7.4	Confusion Matrix	24
7.5	Evaluation Visualisation	25
7.6	Comparison to Baselines	25
7.7	Qualitative Results: Grad-CAM	26
8	Discussion	27
8.1	Interpretation of Overall Results	27
8.2	The Grade 3 Challenge	27
8.3	Overfitting Behaviour	27
8.4	Laser Photocoagulation Scarring: An Unresolved Failure Mode	28
8.5	Comparison with Literature	28
8.6	Clinical Relevance	28
9	Limitations and Ethical Considerations	29
9.1	Technical Limitations	29
9.2	Clinical Limitations	29
9.3	Ethical and Regulatory Considerations	29
9.4	Data Privacy	30
10	Future Work	31
11	Conclusion	32
A	Project Structure	34
B	API Usage Example	35

C Key Source Code Excerpts	36
C.1 Model Definition	36
C.2 CLAHE Preprocessing	36

List of Figures

2.1	Representative fundus photographs, one per <i>DR</i> severity grade (0–4). Grade 0 shows a healthy retina; Grade 4 shows proliferative disease with abnormal neovascularisation.	13
5.1	System architecture. The model achieves a <i>QWK</i> of 0.9053 and 83.6% accuracy on 733 validation images. Training used Adam ($\text{lr} = 10^{-4}$), 15 epochs, batch size 32, weighted CrossEntropyLoss, and ReduceLROnPlateau on a Google Colab Tesla T4 GPU.	17
7.1	Detailed evaluation. Left: normalised confusion matrix. Middle: per-class precision, recall, and F1. Right: sample size versus per-class F1, showing a strong correlation between class prevalence and performance.	25
7.2	Example Grad-CAM heatmap overlaid on a fundus image. Warmer colours (red, orange, yellow) indicate regions that most strongly influenced the prediction. The model attends to the macula and vascular arcades, where diabetic lesions are typically concentrated.	26

List of Tables

2.1	Diabetic retinopathy severity grades used in this project.	12
4.1	Class distribution of the APTOS 2019 dataset.	15
5.1	Training hyperparameters.	19
6.1	Software stack.	21
6.2	HTTP endpoints exposed by the Flask application.	21
7.1	Overall performance on the validation set ($n = 733$).	23
7.2	Per-epoch training and validation metrics.	24
7.3	Per-class performance on the validation set ($n = 733$).	24
7.4	Confusion matrix (row-normalised). Row = actual, column = predicted. .	25
7.5	Comparison with baselines and literature.	25

Chapter 1

Introduction

1.1 Clinical Motivation

Diabetic retinopathy is a microvascular complication of diabetes mellitus and remains one of the leading causes of acquired blindness in adults aged 20–74 [?]. It is estimated that approximately one in three people with diabetes will develop some form of retinopathy during their lifetime, and roughly one in ten will develop a vision-threatening stage [?].

The pathophysiology of *DR* involves progressive damage to the retinal microvasculature driven by chronic hyperglycaemia. It manifests clinically as microaneurysms, haemorrhages, hard exudates and, in advanced stages, neovascularisation [?].

1.2 The Screening Challenge

Regular retinal screening allows *DR* to be detected and treated before vision loss becomes irreversible. However, manual grading of fundus photographs requires trained ophthalmologists. In many regions the ratio of specialists to patients is insufficient to meet screening demand, creating a clear opportunity for computer-aided diagnosis (CAD).

1.3 Problem Statement

Given a single retinal fundus photograph, automatically and reliably classify the severity of diabetic retinopathy into one of five clinically defined grades (0–4), using a deep learning model that can be deployed as a lightweight web service.

1.4 Objectives

- O1.** Design and train a convolutional neural network to classify *DR* severity into five grades.
- O2.** Apply clinically motivated preprocessing (*CLAHE*) to enhance retinal features.

- O3.** Achieve a Quadratic Weighted Kappa comparable to published baselines on the APTOS 2019 dataset.
- O4.** Build a web application for image upload and instant prediction.
- O5.** Provide explainability through Grad-CAM heatmaps.
- O6.** Deploy the system in a reproducible Docker container.

1.5 Contributions

- An end-to-end, reproducible pipeline combining EfficientNet-B3, *CLAHE* preprocessing, and a custom classification head.
- A *QWK* of 0.9053 and accuracy of 83.6% on the APTOS 2019 validation set.
- A containerised Flask web application with a public demo.
- Integration of Grad-CAM explainability into the prediction API.
- A transparent discussion of class imbalance and its effect on Grade 3 performance.

1.6 Report Structure

Chapter 2 provides clinical and technical background. Chapter 3 reviews related work. Chapter 4 describes the dataset. Chapter 5 details the methodology. Chapter 6 covers implementation and deployment. Chapter 7 presents results. Chapter 8 discusses findings. Chapter 9 covers limitations and ethics. Chapter 10 outlines future work. Chapter 11 concludes.

Chapter 2

Clinical and Technical Background

2.1 Diabetic Retinopathy: Pathophysiology

Chronic hyperglycaemia triggers biochemical cascades — activation of the polyol pathway, formation of advanced glycation end-products (AGEs), activation of protein kinase C, and increased oxidative stress — that damage retinal capillaries. The result is pericyte loss, basement membrane thickening, capillary occlusion, and increased vascular permeability, clinically visible as microaneurysms, haemorrhages, and hard exudates. In later stages retinal ischaemia drives VEGF-mediated neovascularisation, the hallmark of proliferative *DR*.

2.2 International Clinical Severity Scale

Table 2.1: Diabetic retinopathy severity grades used in this project.

Grade	Label	Clinical description
0	No DR	No abnormalities.
1	Mild NPDR	Microaneurysms only.
2	Moderate NPDR	More than mild but less than severe: haemorrhages, microaneurysms, hard exudates, or venous beading.
3	Severe NPDR	Extensive haemorrhages in four quadrants, venous beading in two quadrants, or IRMA in one quadrant.
4	Proliferative DR	Neovascularisation, vitreous haemorrhage, or tractional retinal detachment.

2.3 Fundus Imaging

A fundus photograph is acquired with a retinal camera that illuminates the retina through the pupil. Colour fundus photography is the most common modality for *DR* screening:

it is non-invasive, inexpensive, and widely available. Figure 2.1 shows one representative image per severity grade from the APTOS 2019 dataset.

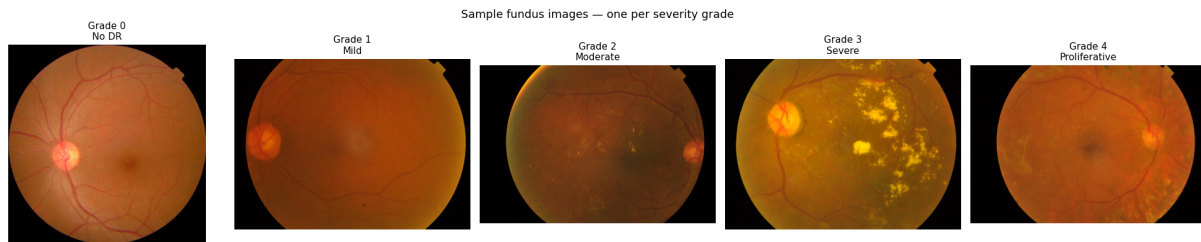


Figure 2.1: Representative fundus photographs, one per DR severity grade (0–4). Grade 0 shows a healthy retina; Grade 4 shows proliferative disease with abnormal neovascularisation.

2.4 Machine Learning for Medical Imaging

Convolutional neural networks dominate medical image classification. Key architectural innovations include residual connections [?], compound scaling in EfficientNet [?], and transfer learning from ImageNet — essential when medical datasets are small. For DR grading, the Quadratic Weighted Kappa is the preferred metric because it penalises large grading errors more heavily than small ones.

Chapter 3

Literature Review

3.1 Early CAD Systems

Early computer-aided diagnosis systems relied on hand-crafted features (vessel segmentation, microaneurysm detection, exudate segmentation). These systems achieved moderate sensitivity but were fragile to variations in illumination, camera, and image quality.

3.2 Deep Learning Era

[?] demonstrated that a deep CNN could achieve sensitivity and specificity comparable to ophthalmologists for referable *DRon* EyePACS. Subsequent studies extended this to multi-grade classification and to other modalities such as OCT.

3.3 APTOS 2019 Competition

The APTOS 2019 Blindness Detection competition on Kaggle attracted over 2,900 teams. Winning solutions used ensembles of EfficientNet, SE-ResNeXt, and Inception architectures with extensive augmentation, test-time augmentation, and 5-fold cross-validation. The top private leaderboard *QWK* was approximately 0.930.

3.4 Explainability in Medical AI

Grad-CAM [?] produces class-discriminative localisation maps by weighting feature maps with the gradients of the target class. In medical imaging, explainability is important for clinician trust and for regulatory approval.

Chapter 4

Dataset

4.1 APTOS 2019 Blindness Detection

The APTOS 2019 dataset consists of 3,662 colour fundus photographs captured under a variety of conditions using different camera models. Each image is labelled with a *DRseverity* grade from 0 to 4.

4.2 Class Distribution

Table 4.1: Class distribution of the APTOS 2019 dataset.

Grade	Label	Count	Proportion
0	No DR	1,805	49.3%
1	Mild NPDR	370	10.1%
2	Moderate NPDR	999	27.3%
3	Severe NPDR	193	5.3%
4	Proliferative	295	8.1%

The dataset is imbalanced: Grade 0 dominates, while Grade 3 is the smallest class with only 5.3% of samples. This imbalance directly impacts per-class performance and is discussed in Chapters 7 and 8.

4.3 Train/Validation Split

The dataset was split 80/20 using stratified sampling by diagnosis grade, with a fixed random seed of 42 for reproducibility:

- Training set: 2,929 images.
- Validation set: 733 images.
- No separate test set; the APTOS competition used a private leaderboard as its test set.

- No cross-validation.

Chapter 5

Methodology

5.1 System Overview

The system pipeline consists of seven stages, illustrated in Figure 5.1:

1. Fundus image acquisition (300×300 RGB input).
2. *CLAHE* preprocessing in LAB colour space.
3. Data augmentation (training only).
4. EfficientNet-B3 backbone feature extraction.
5. Custom classification head.
6. Softmax over 5 classes.
7. Output: DR grade (0–4) with risk score and recommendation.

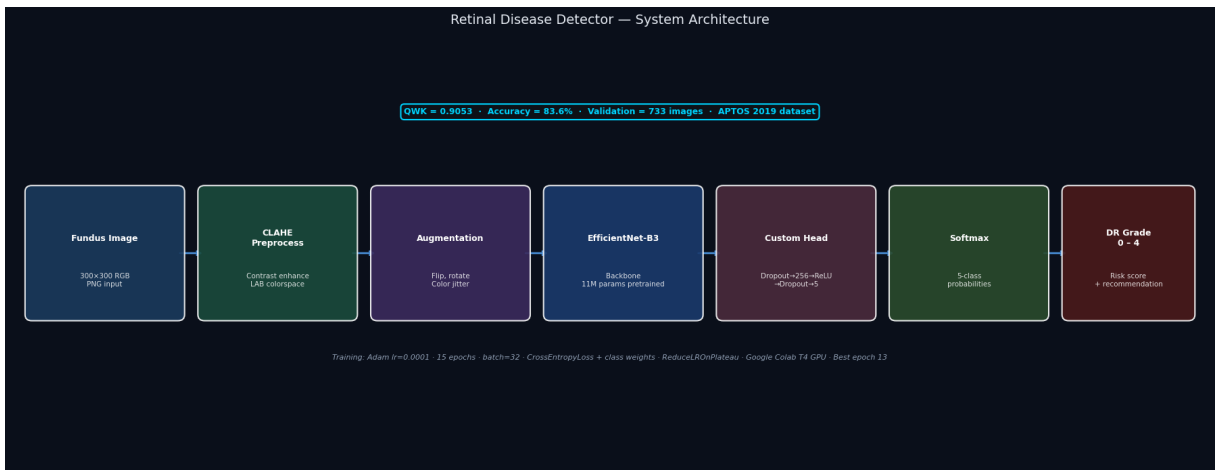


Figure 5.1: System architecture. The model achieves a QWK of 0.9053 and 83.6% accuracy on 733 validation images. Training used Adam ($lr = 10^{-4}$), 15 epochs, batch size 32, weighted CrossEntropyLoss, and ReduceLROnPlateau on a Google Colab Tesla T4 GPU.

5.2 Preprocessing Pipeline

The preprocessing pipeline exactly mirrors the training pipeline:

1. Load image as RGB.
2. Convert to LAB colour space.
3. Apply *CLAHE* on the L channel with `clipLimit=2.0` and `tileGridSize=(8,8)`.
4. Merge channels and convert back to RGB.
5. Resize to 300×300 .
6. Convert to tensor and normalise using ImageNet statistics: $\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$.

5.2.1 Rationale for CLAHE

CLAHE redistributes the histogram of small tiles with a clipping limit. In retinal imaging it increases the visibility of small lesions such as microaneurysms without amplifying noise globally.

5.2.2 Data Augmentation

During training only, the following augmentations were applied: random horizontal and vertical flips, rotation, colour jitter, and zoom.

5.3 Model Architecture

The classifier uses EfficientNet-B3 as a feature extractor, followed by a custom classification head:

$$\begin{aligned}
 \mathbf{z} &= \text{EffNet-B3}(\mathbf{x}), \quad \mathbf{z} \in \mathbb{R}^{1536} \\
 \mathbf{h} &= \text{Dropout}_{0.3}(\mathbf{z}) \\
 \mathbf{h} &= \text{ReLU}(\mathbf{W}_1 \mathbf{h} + \mathbf{b}_1), \quad \mathbf{W}_1 \in \mathbb{R}^{256 \times 1536} \\
 \mathbf{h} &= \text{Dropout}_{0.3}(\mathbf{h}) \\
 \hat{\mathbf{y}} &= \mathbf{W}_2 \mathbf{h} + \mathbf{b}_2, \quad \mathbf{W}_2 \in \mathbb{R}^{5 \times 256}
 \end{aligned} \tag{5.1}$$

The backbone contains approximately 11M parameters and was initialised with ImageNet weights, then fine-tuned end-to-end.

5.4 Training Procedure

Two-phase training was used:

- **Phase 1:** backbone frozen, classification head trained in isolation for the initial epochs to warm up the new layers.
- **Phase 2:** entire network unfrozen and fine-tuned end-to-end with a low learning rate.

Hyperparameters are summarised in Table 5.1.

Table 5.1: Training hyperparameters.

Hyperparameter	Value
Backbone	EfficientNet-B3 (ImageNet pre-trained)
Input size	300×300
Optimizer	Adam
Learning rate	1×10^{-4}
Loss function	CrossEntropyLoss with class weights
Batch size	32
Epochs	15
LR scheduler	ReduceLROnPlateau
Augmentation	Flip, rotation, colour jitter, zoom
Random seed	42
Hardware	NVIDIA Tesla T4 (Google Colab)
Best epoch	13 / 15

5.5 Evaluation Metrics

5.5.1 Quadratic Weighted Kappa

$$\kappa = 1 - \frac{\sum_{i,j} w_{i,j} O_{i,j}}{\sum_{i,j} w_{i,j} E_{i,j}}, \quad w_{i,j} = \frac{(i-j)^2}{(N-1)^2} \quad (5.2)$$

where O is the observed confusion matrix, E the expected confusion matrix under independence, and $N = 5$ the number of grades.

5.5.2 Accuracy, Precision, Recall, F1

Standard definitions were used: Precision = $\frac{TP}{TP+FP}$, Recall = $\frac{TP}{TP+FN}$, $F_1 = \frac{2PR}{P+R}$.

5.6 Explainability: Grad-CAM

Grad-CAM produces a heatmap by computing the gradient of the target class score with respect to the feature maps of the last convolutional layer, then weighting each feature map by the global average of those gradients:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right), \quad \alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_{i,j}^k} \quad (5.3)$$

Chapter 6

Implementation and Deployment

6.1 Software Stack

Table 6.1: Software stack.

Layer	Technology
Model architecture	EfficientNet-B3 (via <code>timm</code> 1.0.3)
Deep learning	PyTorch 2.1, torchvision 0.16
Image processing	OpenCV (headless 4.8.1), Pillow 10.1
Web backend	Flask 3.0
Frontend	HTML, CSS, JavaScript
Deployment	Docker, Gunicorn 21.2, Hugging Face Spaces
Dataset	APTOS 2019 (Kaggle)

6.2 Web Application

Table 6.2: HTTP endpoints exposed by the Flask application.

Route	Method	Purpose
/	GET	Web interface for image upload.
/health	GET	Liveness check and model status.
/debug	POST	Raw logits and probabilities (development only).
/predict	POST	Main prediction endpoint.

The `/predict` endpoint returns a JSON object containing the predicted grade (0–4), human-readable label, clinical description, recommendation, colour code, confidence percentage, per-class probabilities, a Grad-CAM heatmap (base64 PNG), and a low-confidence warning when the maximum softmax probability is below 0.60.

6.3 Docker Container

```
1 FROM python:3.10-slim
2 RUN apt-get update && apt-get install -y \
3     libgl1-mesa-glu libglu1-mesa-dev libxext6 libxrender-dev libgomp1 \
4     && rm -rf /var/lib/apt/lists/*
5 WORKDIR /app
6 COPY requirements.txt .
7 RUN pip install --no-cache-dir -r requirements.txt
8 COPY app/ ./app/
9 COPY data/best_model.pth ./data/best_model.pth
10 EXPOSE 7860
11 CMD ["gunicorn", "--bind", "0.0.0.0:7860", \
12     "--workers", "1", "--timeout", "120", "app.app:app"]
```

Listing 6.1: Dockerfile.

6.4 Deployment

The application is deployed on Hugging Face Spaces using the Docker SDK. Model weights (`best_model.pth`) are tracked via Git LFS and loaded at container startup. The `/health` endpoint supports monitoring.

Chapter 7

Results

7.1 Overall Performance

Table 7.1: Overall performance on the validation set ($n = 733$).

Metric	Value
Quadratic Weighted Kappa (QWK)	0.9053
Accuracy	83.6%
Best training kappa	0.9836
Validation set size	733 images
Training set size	2,929 images
Epochs trained	15
Best epoch	13
Random seed	42

7.2 Training History

Table 7.2 reports per-epoch training and validation metrics. The validation QWK peaked at epoch 13 (0.9053), after which validation loss began to rise while training loss continued to fall — the classic signature of overfitting.

Table 7.2: Per-epoch training and validation metrics.

Epoch	Train Loss	Train Acc	Train κ	Val Loss	Val Acc	Val κ
1	1.3350	0.576	0.5818	1.0128	0.675	0.7930
2	0.8776	0.729	0.8385	0.8256	0.756	0.8717
3	0.7139	0.802	0.8931	0.8159	0.776	0.8968
4	0.5934	0.818	0.9070	0.8297	0.816	0.8946
5	0.5140	0.858	0.9149	0.8489	0.795	0.8965
6	0.4244	0.883	0.9364	0.9484	0.820	0.8971
7	0.3466	0.900	0.9441	1.0289	0.805	0.8854
8	0.2689	0.922	0.9562	1.0473	0.809	0.8778
9	0.2057	0.937	0.9642	1.2117	0.802	0.8690
10	0.2029	0.943	0.9656	1.2268	0.814	0.8874
11	0.1780	0.948	0.9729	1.2420	0.821	0.9017
12	0.1256	0.967	0.9834	1.2970	0.834	0.8980
13	0.1114	0.968	0.9836	1.4430	0.836	0.9053
14	0.1196	0.965	0.9808	1.4192	0.819	0.8912
15	0.1317	0.965	0.9794	1.5145	0.831	0.9023

7.3 Per-Class Performance

Table 7.3: Per-class performance on the validation set ($n = 733$).

Grade	Label	Precision	Recall	F1	Support
0	No DR	0.99	0.98	0.98	361
1	Mild NPDR	0.64	0.61	0.62	74
2	Moderate NPDR	0.77	0.82	0.80	200
3	Severe NPDR	0.39	0.33	0.36	39
4	Proliferative	0.66	0.64	0.65	59

7.4 Confusion Matrix

Table 7.4 shows the row-normalised confusion matrix. The matrix is asymmetric in a clinically informative way: when the model misclassifies a Grade 3 image, it most often predicts Grade 2 (38% of cases) or Grade 4 (26%) — adjacent, clinically related stages — rather than Grade 0.

Table 7.4: Confusion matrix (row-normalised). Row = actual, column = predicted.

	G0	G1	G2	G3	G4	Support
G0	352 (98%)	8 (2%)	1 (0%)	0 (0%)	0 (0%)	361
G1	4 (5%)	45 (61%)	21 (28%)	1 (1%)	3 (5%)	74
G2	0 (0%)	7 (4%)	165 (82%)	15 (8%)	13 (7%)	200
G3	0 (0%)	1 (3%)	15 (38%)	13 (33%)	10 (26%)	39
G4	0 (0%)	0 (0%)	13 (22%)	8 (14%)	38 (64%)	59

7.5 Evaluation Visualisation



Figure 7.1: Detailed evaluation. Left: normalised confusion matrix. Middle: per-class precision, recall, and F1. Right: sample size versus per-class F1, showing a strong correlation between class prevalence and performance.

7.6 Comparison to Baselines

Table 7.5: Comparison with baselines and literature.

Model	QWK
Random baseline	0.000
Simple CNN	~0.700
EfficientNet-B3 (this work)	0.9053
APTOS 2019 competition winner	~0.930

7.7 Qualitative Results: Grad-CAM

Figure 7.2 shows a Grad-CAM overlay for a Moderate NPDR prediction. The heatmap concentrates on the posterior pole — the region containing the macula, the optic disc, and the major vascular arcades — which is where exudates, haemorrhages, and microaneurysms are most dense in this image. The model’s attention is well aligned with the clinically relevant retinal anatomy, supporting the plausibility of its decisions.

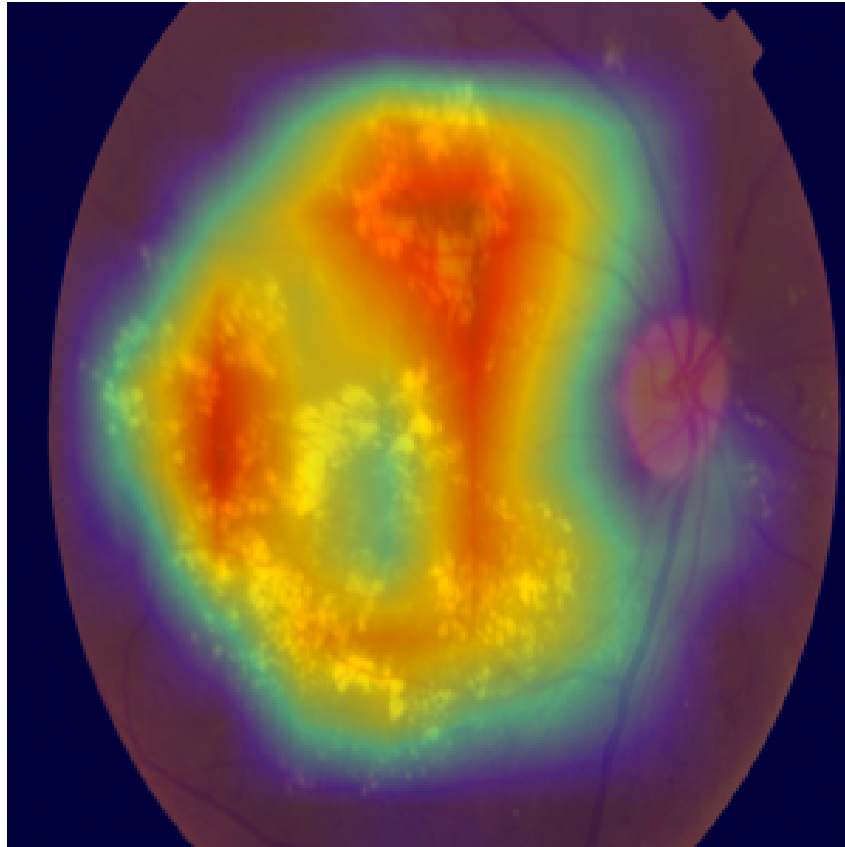


Figure 7.2: Example Grad-CAM heatmap overlaid on a fundus image. Warmer colours (red, orange, yellow) indicate regions that most strongly influenced the prediction. The model attends to the macula and vascular arcades, where diabetic lesions are typically concentrated.

Chapter 8

Discussion

8.1 Interpretation of Overall Results

The QWK of 0.9053 reflects strong agreement between the model and expert labels, approaching the APTOS 2019 competition winner (0.930) with a single architecture and a single split. The high F1 on Grade 0 (0.98) reflects the fact that healthy retinas are visually distinct and well represented in the dataset.

8.2 The Grade 3 Challenge

Grade 3 (Severe NPDR) is the weakest class ($F1 = 0.36$). This is directly attributable to its small sample size: only 39 validation images and 193 in the full dataset. The scatter plot in Figure 7.1 (right panel) shows a clear monotonic relationship between class prevalence and F1, confirming this is a *data limitation*, not an architectural failure.

Three complementary strategies would address this gap:

1. Collect more Grade 3 examples.
2. Apply targeted augmentation to Grade 3.
3. Use stronger class-weighted or focal loss.

8.3 Overfitting Behaviour

Validation loss rises steadily after approximately epoch 6, while training loss continues to fall. Validation QWK nevertheless peaks at epoch 13, suggesting that the model is still learning class-relevant features even as calibration degrades. This pattern is common with small medical datasets and could be mitigated with early stopping, stronger regularisation, or additional data.

8.4 Laser Photocoagulation Scarring: An Unresolved Failure Mode

A clinically important limitation not captured by the APTOS labels is that the dataset does not encode treatment history. Eyes that have undergone prior panretinal laser photocoagulation carry a characteristic grid of small, deliberately placed pale lesions. These scars can be visually similar to hard exudates and are therefore graded as active Moderate to Severe *NPDR*. The model inherits this ambiguity: it has learned the visual signature of exudate-like lesions without any signal to distinguish treated from untreated retina. This is a known, unresolved limitation of the current model, and would need explicit treatment labels to be corrected.

8.5 Comparison with Literature

The result is competitive with published single-model baselines on APTOS 2019. Ensembles typically achieve higher *QWK*, but the simplicity of this pipeline — one architecture, one split, no test-time augmentation — makes the result easy to reproduce and extend.

8.6 Clinical Relevance

The system could serve as a triage tool in screening workflows, flagging images that warrant ophthalmologist review. However, the low F1 on Grade 3 means the system cannot currently be relied on to detect severe non-proliferative *DR*, which is exactly the stage where timely intervention matters most. This reinforces the importance of addressing class imbalance before any clinical deployment.

Chapter 9

Limitations and Ethical Considerations

9.1 Technical Limitations

- Single dataset (APTOS 2019); no external validation.
- Class imbalance, especially for Grade 3.
- No separate held-out test set.
- No cross-validation; a single fixed split is reported.
- CPU-only inference in production (no GPU optimisation).
- Grad-CAM roughly doubles inference cost.
- Overfitting after epoch 13.

9.2 Clinical Limitations

- Not validated on prospectively collected clinical data.
- No patient metadata used (age, diabetes duration, HbA1c).
- No comparison against ophthalmologist performance in a reader study.
- Performance across different cameras and populations is unknown.
- Cannot distinguish post-treatment laser scarring from active exudates.

9.3 Ethical and Regulatory Considerations

- This tool is **not** a medical device and has not received FDA, CE, or equivalent approval.
- It must not be used for diagnosis, treatment, or clinical decision-making.
- Patient privacy must be protected: uploaded images should not be stored or logged without explicit consent.

- Dataset bias may affect performance across ethnic groups, camera models, and clinical settings.
- Any future clinical use would require regulatory approval, prospective trials, and clinical validation.

9.4 Data Privacy

Fundus images are personally identifiable medical data. The public demo should be used only with images the user is authorised to share. The `/debug` endpoint should be disabled in production because it exposes internal model state.

Chapter 10

Future Work

1. **Data:** collect more Grade 3 examples; apply targeted augmentation; use stronger class-weighted or focal loss.
2. **Validation:** k -fold cross-validation and external validation on Messidor, EyePACS, and IDRiD.
3. **Modelling:** EfficientNetV2, ConvNeXt, Vision Transformers, and ensembles.
4. **Treatment awareness:** incorporate treatment history labels to distinguish laser scarring from active exudates.
5. **Explainability:** lesion segmentation and uncertainty estimation (MC dropout, deep ensembles).
6. **Deployment:** authentication, rate limiting, non-root Docker user, GPU inference.
7. **Clinical:** reader study with ophthalmologists; regulatory pathway (CE / FDA).

Chapter 11

Conclusion

This project demonstrates a complete, reproducible deep learning pipeline for diabetic retinopathy severity classification. Using EfficientNet-B3 with *CLAHE* preprocessing and a class-weighted loss, the model achieves a Quadratic Weighted Kappa of 0.9053 and 83.6% accuracy on the APTOS 2019 validation set. The system is deployed as a containerised web service with Grad-CAM explainability. Grade 3 performance remains limited by class imbalance and is compounded by the dataset’s lack of treatment labels. Future work should address both gaps through targeted data collection, loss re-weighting, and treatment-aware labelling. The system is a research prototype, not a medical device, and would require clinical validation and regulatory approval before any real-world diagnostic use.

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Appendix A

Project Structure

```
1 retinal-disease-detector/
2 |-- app/
3 |   |-- app.py
4 |   |-- static/
5 |   '-- templates/
6 |       '-- index.html
7 |-- model/
8 |   |-- dataset.py
9 |   |-- model.py
10 |   |-- train.py
11 |   '-- evaluate.py
12 |-- notebooks/
13 |   |-- 01_data_exploration.ipynb
14 |   '-- 02_model_training.ipynb
15 |-- docs/SPEC.md
16 |-- Dockerfile
17 |-- requirements.txt
18 '-- README.md
```

Appendix B

API Usage Example

```
1 curl -X POST http://localhost:7860/predict \  
2     -F "file=@fundus.jpg"
```

Appendix C

Key Source Code Excerpts

C.1 Model Definition

```
1 class RetinalClassifier(nn.Module):
2     def __init__(self, num_classes=5, dropout=0.3):
3         super().__init__()
4         self.backbone = timm.create_model(
5             "efficientnet_b3", pretrained=False, num_classes=0
6         )
7         self.classifier = nn.Sequential(
8             nn.Dropout(dropout),
9             nn.Linear(self.backbone.num_features, 256),
10            nn.ReLU(),
11            nn.Dropout(dropout),
12            nn.Linear(256, num_classes),
13        )
14
15    def forward(self, x):
16        return self.classifier(self.backbone(x))
```

Listing C.1: RetinalClassifier.

C.2 CLAHE Preprocessing

```
1 def _apply_clahe(img_rgb: np.ndarray) -> np.ndarray:
2     lab = cv2.cvtColor(img_rgb, cv2.COLOR_RGB2LAB)
3     l, a, b = cv2.split(lab)
4     clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
5     l = clahe.apply(l)
6     return cv2.cvtColor(cv2.merge([l, a, b]), cv2.COLOR_LAB2RGB)
```

Listing C.2: CLAHE preprocessing.